Integrale curbilinii

$$\int_\gamma P(x,y,z)\,ds = \int_a^b P(x(t),y(t),z(t))\cdot\sqrt{x'^2(t)+y'^2(t)+z'^2(t)}\,dt$$

integ. curbil. de speța I

$$L(\gamma)=\int_\gamma ds \quad \text{– lungimea curbei } \gamma$$

$$\gamma:\begin{cases} x=x(t)\\ y=y(t)\\ z=z(t)\end{cases} \quad t\in[a,b]$$

$$\int_\gamma P\,dx+Q\,dy+R\,dz = \int_a^b \big[P(x(t),y(t),z(t))\,x'(t)+Q(x(t),y(t),z(t))\cdot y'(t)+R(x(t),y(t),z(t))\cdot z'(t)\big]\,dt$$

integ. curbil. de speța a II-a

$\bar F = P\bar i + Q\bar j + R\cdot\bar k$ câmp de forțe

$L=\int_\gamma \bar F\,d\bar r = \int_\gamma P\,dx+Q\,dy+R\,dz$

lucru mecanic

$\bar r = x\bar i+y\bar j+z\bar k$

$d\bar r = dx\bar i+dy\bar j+dz\bar k$

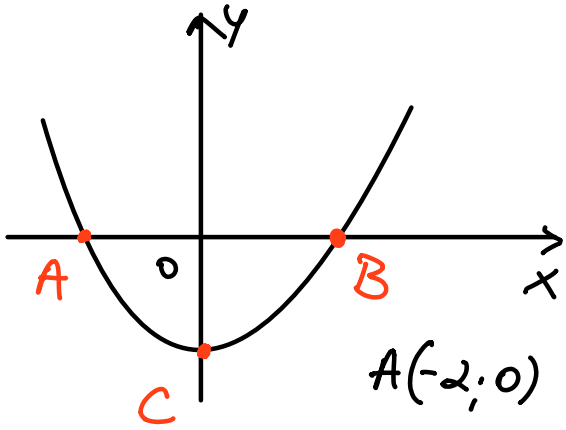
$\bar V = P\bar i+Q\bar j+R\cdot\bar k$ câmp vectorial

$C=\int_\gamma \bar V\,d\bar r = \int P\,dx+Q\,dy+R\,dz$

circulația câmp. vect. $\bar V$ de-a lungul curbei γ

① $\int_{\overset{\frown}{ACB}} \frac{dx+dy}{y+5}$; $y=x^2-4$; A, B fiind pct. unde curba taie Ox
C pct. cel mai de jos

$$\int_{\overset{\frown}{ACB}} \frac{dx+dy}{y+5} = \int_{\overset{\frown}{ACB}} \frac{dx}{y+5} + \frac{dy}{y+5} \overset{\text{Not}}{=} J$$



$A(-2;0)$
$B(2;0)$

$x = t$
$y = t^2-4$
$\xrightarrow{t\in[-2;2]}$ $dx = dt$
$dy = 2t\,dt$

$$J = \int_{-2}^{2}\left(\frac{1}{t^2+1} + \frac{2t}{t^2+1}\right)dt = \int_{-2}^{2}\frac{1}{t^2+1}dt + \int_{5}^{5}\frac{1}{g}dg$$

Notăm $t^2+1=g$ $\quad$ $t=-2 \Rightarrow g=5$
$2t\,dt = dg$ $\quad$ $t=2 \Rightarrow g=5$

$$J = \operatorname{arctg} t\Big|_{-2}^{2} + \ln(g)\Big|_{5}^{5} = \operatorname{arctg} 2 - \operatorname{arctg}(-2) = 2\operatorname{arctg} 2$$

② $x^2+y^2+z^2=1$ $\quad z\in[0,2]$
$x^2+y^2=\frac{1}{4}$

$\frac{1}{4}+z^2=1 \Rightarrow z^2=\frac{3}{4} \Rightarrow z=\pm\frac{\sqrt{3}}{2}$ $\Big|_{z\in[0,2]} \Rightarrow z=\frac{\sqrt{3}}{2}$

$$\begin{cases} x^2+y^2=\frac{1}{4} \\ z=\frac{\sqrt{3}}{2} \end{cases}$$

$(x-x_0)^2+(y-y_0)^2=R^2$

$$\begin{cases} x=x_0+R\cos t \\ y=y_0+R\sin t \end{cases} ; t\in[0,2\pi)$$

$$\Rightarrow \begin{cases} x=\frac{1}{2}\cos t \\ y=\frac{1}{2}\sin t \\ z=\frac{\sqrt{3}}{2} \end{cases} ; t\in[0,2\pi)$$

$$\int_{\mathcal{C}} ds = \int_0^{2\pi} \sqrt{\frac{1}{4}\sin^2 t + \frac{1}{4}\cos^2 t + 0}\, dt = \int_0^{2\pi} \frac{1}{2}\cdot 1\, dt = \frac{1}{2}t\Big|_0^{2\pi}$$

$$= \frac{2\pi - 0}{2} = \pi$$

3 $L = ?$ $\quad \vec{F} = \frac{1}{x^3+y^3}\vec{i}$ $\quad \mathcal{C}: x^2+y^2 = a^2 \,;\, x, y \geq 0$

$$\begin{cases} x = a\cos t \\ y = a\sin t \end{cases} \quad t \in \left[0, \frac{\pi}{2}\right)$$

$$L = \int_{\mathcal{C}} \vec{F}\, d\vec{r} = \int_{\mathcal{C}} \frac{1}{x^3+y^3}\, dx = \int_0^{\frac{\pi}{2}} \frac{-a\sin t}{a^3\cos^3 t + a^3\sin^3 t}\, dt$$

$$x = a\cos t$$
$$dx = -a\sin t\, dt$$

$$L = \frac{-1}{a^2}\int_0^{\pi/2} \frac{\sin t}{\cos^3 t + \sin^3 t}\, dt$$

Notăm $\operatorname{tg} t = g \Rightarrow t = \operatorname{arctg} g$

$$dt = \frac{1}{1+g^2}\, dg$$

$$\sin^2 t = \frac{g^2}{1+g^2} \Rightarrow \sin t = \frac{g}{\sqrt{1+g^2}} \,;\, \cos t = \frac{1}{\sqrt{1+g^2}}$$

$$\Rightarrow L = \frac{-1}{a^2}\int_0^{\infty} \frac{\frac{g}{\sqrt{1+g^2}}}{\frac{1}{(1+g^2)\sqrt{1+g^2}} + \frac{g^2\cdot g}{(1+g^2)\sqrt{1+g^2}}} \cdot \frac{1}{1+g^2}\, dg$$

$$L = -\frac{1}{a^2}\int_0^{\infty} \frac{q}{1+q^3}\cdot dq$$

$q^3 = u \Rightarrow q = u^{1/3}$

$dq = \frac{1}{3}u^{-2/3}du$

$$\Rightarrow L = -\frac{1}{a^2}\int_0^{\infty} \frac{u^{1/3}}{1+u}\cdot\frac{1}{3}u^{-2/3}du$$

$$L = -\frac{1}{3a^2}\int_0^{\infty} \frac{u^{-1/3}}{1+u}du \Rightarrow$$

$p-1 = -1/3 \Rightarrow p = 2/3$

$p+q = 1 \Rightarrow q = 1/3$

$$\Rightarrow L = -\frac{1}{3a^2}\cdot B(2/3, 1/3) = -\frac{1}{3a^2}\cdot\frac{\Gamma(2/3)\cdot\Gamma(1/3)}{\Gamma(1)}$$

$$\Rightarrow L = -\frac{1}{3a^2}\cdot\frac{\pi}{\sin(\frac{1}{3}\pi)} = -\frac{1}{3a^2}\cdot\frac{\pi\cdot 2}{\sqrt{3}} = -\frac{2\pi\sqrt{3}}{9a^2}$$

(4) Să se calculeze circulația câmpului vectorial $\bar{v} = x\bar{i} + (y-z)\bar{j} + z\bar{k}$ de-a lungul segmentului $[AB]$

$A(-1,1,2)$; $B(2,0,3)$. $C = ?$

$$\frac{x-(-1)}{2-(-1)} = \frac{y-1}{0-1} = \frac{z-2}{3-2} = t \Rightarrow \frac{x+1}{3} = 1-y = z-2 = t$$

$$\Rightarrow \begin{cases} x+1 = 3t \\ y = 1-t \\ z = t+2 \end{cases} \Rightarrow \begin{cases} x = 3t-1 \\ y = 1-t \\ z = t+2 \end{cases}$$

$x \in [-1;2]$

$x+1 \in [0;3] \Rightarrow t \in [0,1]$

$$C = \int_{AB} x\,dx + (y-z)\,dy + z\,dz$$

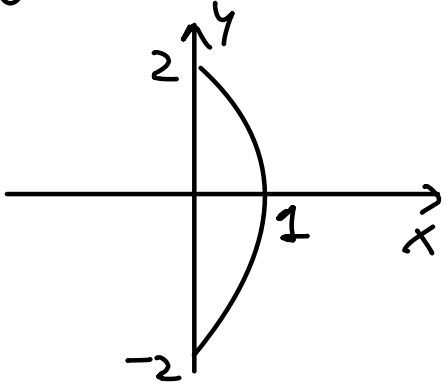
$$\begin{cases} x = 3t-1 \\ y = 1-t \\ z = t+2 \end{cases} \longrightarrow \begin{cases} dx = 3dt \\ dy = -dt \\ dz = dt \end{cases}$$

$$C = \int_0^1 [(3t-1)\cdot 3 + (1-t-t-2)(-1) + (t+2)]\,dt$$

$$= \int_0^1 [9t - 3 - (-1-2t) + t + 2]\,dt = \int_0^1 [10t - 1 + 1 + 2t]\,dt$$

$$C = \int_0^1 12t\,dt = 12\,\frac{t^2}{2}\Big|_0^1 = 6(1-0) = 6$$

5 Să se calculeze circulația câmpului vectorial $\bar{v} = \sqrt{1-x^2}\,\bar{i} + x\bar{j}$, când $\mathcal{C}: x^2 + \frac{y^2}{4} = 1 \quad x \geq 0$



$$C = \int_{\mathcal{C}} \sqrt{1-x^2}\,dx + x\,dy$$

$$\begin{cases} x = \cos t \\ y = 2\sin t \end{cases} \quad t \in \left[-\frac{\pi}{2}; \frac{\pi}{2}\right]$$

$$\longrightarrow \begin{cases} dx = -\sin t\,dt \\ dy = 2\cos t\,dt \end{cases}$$

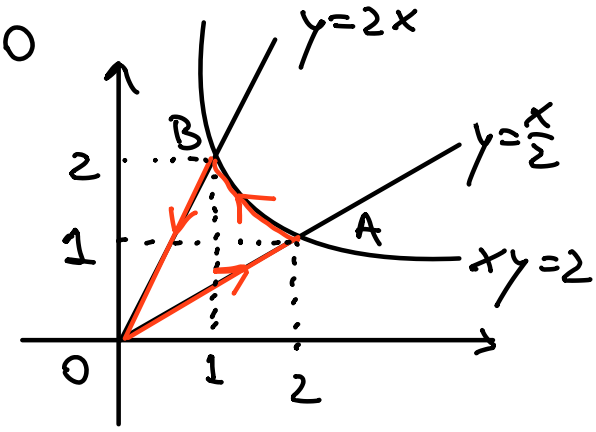
$$C = \int_{-\frac{\pi}{2}}^{\pi/2} \sqrt{1-\cos^2 t}\cdot(-\sin t) + \cos t \cdot 2\cos t\,dt$$

$$C = \int_{-\pi/2}^{\pi/2} |\sin t|(-\sin t) + 1 + \cos 2t\,dt$$

$$C = \underbrace{\int_{-\pi/2}^{0} \sin^2 t\,dt + \int_0^{\pi/2} -\sin^2 t\,dt}_{0} + \int_{-\pi/2}^{\pi/2} (1+\cos 2t)\,dt =$$

$$C = 0 + t\Big|_{-\pi/2}^{\pi/2} + \frac{-\sin 2t}{2}\Big|_{-\pi/2}^{\pi/2} = \pi - \underbrace{\frac{\sin\pi}{2}}_{0} + \underbrace{\frac{\sin(-\pi)}{2}}_{0} = \pi$$

⑥ Să se calculeze circulația câmpului vectorial $\bar{v} = (x+y)\bar{i} - y\bar{j}$, când δ este intersecția mărginirilor curbe: $xy = 2$; $y = 2x$; $y = \frac{x}{2}$ $\quad x \geq 0$



$y = \frac{2}{x}$ | $\Rightarrow \frac{2}{x} = 2x \Rightarrow x^2 = 1 \Rightarrow x = 1$

$y = 2x$ | $y = 2$

$y = \frac{x}{2} \Rightarrow \frac{2}{x} = \frac{x}{2} \Rightarrow x^2 = 4 \Rightarrow x = 2$

$y = 1$

$$C = \int_{\delta} (x+y)\,dx + (-y)\,dy =$$

$$C = \int_{[OA]} (x+y)\,dx - y\,dy + \int_{\overset{\frown}{AB}} (x+y)\,dx - y\,dy + \int_{[BO]} (x+y)\,dx - y\,dy$$

$[OA]$: $\begin{cases} x = t \\ y = t/2 \end{cases} \longrightarrow \begin{cases} dx = dt \\ dy = 2dt \end{cases}$

$x \in [0,2] \quad t \in [0,2]$

$\overset{\frown}{BA}$: $\begin{cases} x = t \\ y = \frac{2}{t} \end{cases} \longrightarrow \begin{cases} dx = dt \\ dy = 2\frac{-1}{t^2}dt \end{cases}$

$x \in [1;2] \quad t \in [1;2]$

$[OB]$ $\begin{cases} x = t \\ y = 2t \end{cases} \longrightarrow \begin{cases} dx = dt \\ dy = \frac{dt}{2} \end{cases}$

$x \in [0,1] \quad t \in [0,1]$

$$C=\int_0^2\left(\left(t+\frac{t}{2}\right)\cdot 1-\frac{t}{2}\cdot\frac{1}{2}\right)dt-\int_1^2\left(t+\frac{2}{t}\right)-\frac{2}{t}\cdot\left(\frac{-2}{t^2}\right)dt-\int_0^1 3t-2t\cdot 2\,dt$$

$$C=\int_0^2\frac{3t}{2}-\frac{t}{4}\,dt-\int_1^2 t+\frac{2}{t}+\frac{4}{t^3}\,dt-\int_0^1 -t\,dt$$

$$C=\left(\frac{3}{2}\cdot\frac{t^2}{2}-\frac{1}{4}\cdot\frac{t^2}{2}\right)\Big|_0^2-\left(\frac{t^2}{2}+2\ln(t)+\frac{2}{4}\cdot\frac{t^{-2}}{-2}\right)\Big|_1^2+\frac{t^2}{2}\Big|_0^1$$

$$C=\frac{3}{4}(4-0)-\frac{1}{8}(4-0)-\left[\frac{4-1}{2}+2\cdot\ln 2-2\ln 1\right.$$

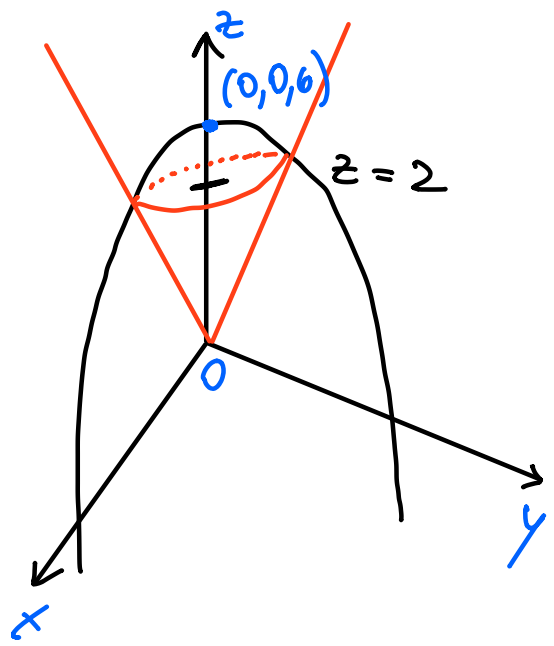
$$\left.-2\left(\frac{1}{4}-1\right)\right]+\frac{1-0}{2}$$

$$C=\frac{3}{4}\cdot 4-\frac{1}{8}\cdot 4-\left[\frac{3}{2}+2\ln 2-2\left(\frac{-3}{4}\right)\right]+\frac{1}{2}$$

$$C=3-\frac{1}{2}-\left[\frac{3}{2}+2\ln 2+\frac{3}{2}\right]+\frac{1}{2}=3-\left[3+2\ln 2\right]$$

$$C=-2\ln 2$$

(7) Să se calculeze circulația câmpului vectorial $\bar{v}=y\bar{i}+z^2\bar{j}+x\bar{k}$, când δ este intersecția următoarelor suprafețe: $\delta:\begin{cases}z=\sqrt{x^2+y^2}\\ z=6-(x^2+y^2)\end{cases}$



$$z=6-z^2\Rightarrow z^2+z-6=0$$

$$\Delta=1+4\cdot 1\cdot 6=25\rightarrow z_{1,2}=\frac{-1\pm 5}{2}$$

$$z_1=2;\ z_2=-3$$

$$x^2+y^2=4\Rightarrow\begin{cases}x=2\cos t\\ y=2\sin t\\ z=2\end{cases}\qquad x\in[0;2\pi)$$

$$C = \int_{\gamma} y\,dx + z^2\,dy + x\,dz =$$

$$C = \int_0^{2\pi} 2\sin t \cdot 2(-\sin t) + 4 \cdot 2 \cdot \cos t + 2\cos t \cdot 0 \; dt$$

$$C = \int_0^{2\pi} -4\sin^2 t + 8\cos t \, dt = \int_0^{2\pi} -2(1-\cos 2t) + 8\cos t \; dt$$

$$C = \int_0^{2\pi} -2 + 2\cos 2t + 8\cos t \, dt$$

$$C = -2t\Big|_0^{2\pi} + \cancel{2} \cdot \frac{\sin 2t}{\cancel{2}}\Big|_0^{2\pi} + 8 \cdot \sin t\Big|_0^{2\pi}$$

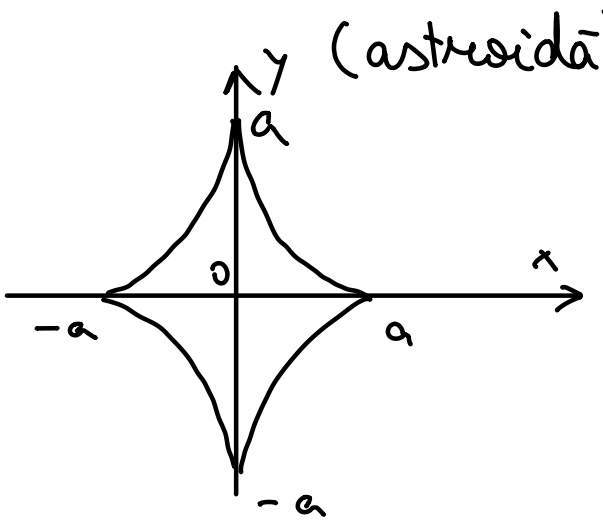
$$C = -2(2\pi - 0) + \underbrace{\sin(4\pi)}_{0} - \underbrace{\sin(0)}_{0} + 8 \cdot \underbrace{\sin(2\pi)}_{0} - \underbrace{\sin 0}_{0}$$

$$C = -4\pi$$

⑧ Aria regiunii limitată de curba $\sqrt[3]{x^2} + \sqrt[3]{y^2} = \sqrt[3]{a^2}$ (astroidă)

$$A = \frac{1}{2}\int_{\gamma} x\,dy - y\,dx$$

$$\begin{cases} x = a\cos^3 t \\ y = a\sin^3 t \end{cases} \quad t \in [0, 2\pi)$$



$$A = \frac{1}{2}\int_0^{2\pi} a\cos^3 t \cdot a \cdot 3 \cdot \sin^2 t \cos t - a\sin^3 t \cdot a\, 3\cos^2 t(-\sin t)\,dt$$

$$A = \frac{3a^2}{2}\int_0^{2\pi} \cos^2 t \sin^2 t(\cos^2 t + \sin^2 t)\,dt = \frac{3a^2}{2}\int_0^{2\pi} \sin^2 t \cos^2 t\,dt$$

$2\sin x \cos x = \sin 2x$

$$A = \frac{3a^2}{2}\cdot\frac{1}{4}\int_0^{2\pi} \sin^2 2t\,dt = \frac{3a^2}{8}\int_0^{2\pi} \frac{1-\cos 4t}{2}\,dt$$

$$A = \frac{3a^2}{8}\cdot\frac{1}{2}\left(t - \frac{\sin 4t}{4}\right)\Big|_0^{2\pi} = \frac{3a^2}{16}\left[2\pi - 0 - \frac{\sin(8\pi) - \sin 0}{4}\right]$$

$$A = \frac{3a^2}{16}\left(2\pi - \frac{0-0}{4}\right) = \frac{3a^2}{16}\cdot 2\pi = \frac{3a^2\pi}{8}$$